DASSOUNDO ADEBAYO SERGIO

COMPUTER SCIENCE ENGINEER
SPECIALIZING IN ARTIFICIAL INTELLIGENCE & DATA,
EMBEDDED SOFTWARE AND CYBERSECURITY
Apprentice AI / RAG Engineer – FastAPI at Yieloo · Master of Science Data & AI – Epitech

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France & Canada
My LinkedIn profile
My web site

SUMMARY

Engineer and researcher combining skills in data science and artificial intelligence (generative AI, RAG, agents and multi-agent systems, LLMs), embedded software, cybersecurity and software development, with academic and industrial experience and a strong focus on technological innovation.

KEY SKILLS

- Generative AI & agents: RAG and agentic RAG (query reformulation, LLM-judge "faithfulness" scoring), multi-agent systems, advanced prompt engineering
- LangChain / LangGraph, NLP, Computer Vision, multi-provider LLMs (Groq/Llama 3, Gemini 2.5 Flash, GPT-5, DeepSeek, Mistral Large, Codestral, Kimi K2.6)
- Speech-to-Text / Text-to-Speech (Whisper, OpenAI TTS), real-time search agents (Tavily API, GitHub Code Search API)
- Vector databases (pgvector) & Data Engineering (SQL, PostgreSQL, Supabase: Auth, Row Level Security)
- Backend & API: FastAPI + Pydantic, REST API, Next.js (App Router), schema validation (Zod)
- Application security: OAuth2, AES-256-GCM secret encryption, distributed rate limiting (Upstash Redis), multi-tenant architecture
- Development: C / C++ / Java / Python / TypeScript / VHDL / SQL/ PHP/ HTML/ ROS2
- Automated validation & testing (Robot Framework, JMeter)
- Linux (Debian, Ubuntu) & continuous integration (Git / GitHub / GitLab CI/CD, Vercel)
- Cybersecurity & secure protocols (TLS, SSH)
- Simulation & analysis: MATLAB / Simulink / LTspice / Vivado
- Electronic design & microcontrollers (STM32 Cube, Arduino, ESP32, Raspberry Pi), CAD (Altium Designer, SolidWorks)
- Project management & collaborative work (Agile / Scrum)
- Embedded systems, Real-time programming
- Software engineering
- UML modeling & specifications

CERTIFICATIONS

- Certified Masterclass on Artificial Intelligence
- Cybersecurity certification
- Data Protection Certification
- Project Management MOOC
- Python Desktop Application Development (Full GUI Stack)
- Prevention_Sup
- CLOE

PROFESSIONAL EXPERIENCE

AI / RAG Engineer - FastAPI – **France** (Ongoing) *2026 - 2028*
YIELOO

Generative AI Engineer (Freelance / Co-founder) - **France** *2026*
DASIA

Chassis SW Development & Integration Engineer - **France** *2024 - 2026*
Renault Group

Research internship in embedded systems & AI - Université de Sherbrooke - **Canada** *2025*
Université de Sherbrooke

Final-year engineering project - **France** *2025 - 2026*
Thales / Eseo

President of the Local Youth Foundation - **Benin** *2020 - 2023*
ONG JPA

Intern in mechanical manufacturing on machine tools - **Benin** *2019*
Centre de perfectionnement des personnes d'entreprises

MAIN TECHNICAL AND PROFESSIONAL PROJECTS

AI / RAG Engineer – FastAPI (Company)

2026 - 2028 : ***YIELOO – software publisher, airport sector - France***

- Member of Yieloo's AI committee: defining the strategy for integrating AI into the business tools of a software publisher for the airport sector (aeronautics, testing and calibration laboratories).
- Assessment of technical and regulatory risks and identification of use cases to automate, in view of a local AI solution tailored to each client's needs (airports, laboratories).
- Organizing and leading client meetings (airports, industry stakeholders) to scope tailor-made solutions that support compliance with aeronautical standards.
- Development and calibration of AI models (RAG architecture, FastAPI) meeting the internal needs of Yieloo and those of its clients, beyond business-task automation.

Multi-agent Generative AI Engineer (Start-up)

2026 : ***DASIA*** *-* ***France***

- Building and deploying a multi-agent platform (6 specialized agents: General, HR, Finance/Legal, Operations/Sales, Embedded Systems, Data Extraction) with an end-to-end RAG pipeline.
- Development of an intelligent multi-LLM router across 7 models (Groq/Llama 3, Gemini 2.5 Flash, GPT-5, DeepSeek, Mistral Large, Codestral, Kimi K2.6) based on query type and target latency.
- Design of an agentic RAG agent (LangGraph + LangChain, exposed via FastAPI): automatic query reformulation and verification of answer faithfulness to sources (faithfulness scoring).
- Integration of autonomous real-time search agents (Tavily API, GitHub Code Search API) and voice capabilities (Whisper, OpenAI TTS).

Software Development & Integration Engineer – Chassis SW, Vehicle Prototype (Company)

2024 - 2026 : ***Renault Group - France***

- Porting of chassis control software to ROS2/NUC.
- Automation of software build and deployment (CI/CD with GitLab).
- Development of a hybrid dSPACE – ROS2 solution.
- Design and testing of human–machine interfaces (HMI).
- Validation and optimization of algorithms on prototypes.
- Algorithm testing on prototype vehicles.

Université de Sherbrooke (Laboratory)

2025 : ***Université de Sherbrooke - Canada***

- Programming of a software interface for bioreactor control.
- Automation of the perfusion and decellularization system.
- Integration of the reactor's sensors, pumps and valves.
- Embedded code optimization, functional validation of the system and deployment of AI models.
- Contribution to the specifications and to component selection.

Final-year engineering project (Company - School)

2025 – 2026 : ***Thales - France***

- Software architecture design
- Development of search and exploration algorithms in unknown environments.
- UML modeling: sequence, object and class diagrams.
- C++ / Python programming for simulation and robotic control.
- Testing and validation of autonomous behaviors in simulation.

Engineering projects (School)

2023 - 2026 : ***ESEO – France***

- Software development in C (vehicle fleet management) and Java (game application).
- Design and programming of embedded systems and Bluetooth remote-controlled cars.
- Electronic detection projects and printed circuit board design (Altium Designer).
- 3D modeling of technical systems with SolidWorks.

Electronics & connected devices (Company)

2022 : ***Digital-Tech*** *- Benin*

- Design of power supply boards (switching and linear)
- Microcontroller programming (C/C++)
- Implementation of connectivity (UART, SPI, I2C, CAN)
- Testing and validation of electronic circuits
- Integration of modules into an embedded system

Manufacturing and industrial maintenance (Company)

2019 : ***CPPE*** *- Benin*

- Manufacturing and machining of mechanical parts
- Preventive maintenance of equipment
- Blueprint reading and compliance checks
- Operation of industrial machine tools
- Welding and metal assembly work

MY MAIN RESEARCH PROJECTS

LLM multi-agent system for decision support in regulatory compliance

2026 : France – Personal applied research project (ongoing)

- Design of an autonomous LLM-based multi-agent system (specialized agents: standards, risk analysis, decision synthesis) to support decision-making in regulated environments (critical infrastructure, airport sector).
- Study of coordination, planning and arbitration between collaborative agents: task allocation, conflict resolution, escalation to a human.
- Implementation of graph-based agent orchestration (LangGraph) with shared memory and RAG over normative corpora (aviation, quality).
- Experimental performance evaluation: relevance, source faithfulness (faithfulness scoring), decision time, error rate compared with a single-agent analysis.
- Full decision traceability (logging of inter-agent exchanges and sources used) for auditability.

Local, reliable and sovereign AI architecture for critical infrastructure

2026 : France – Personal applied research project (ongoing)

- Design of a generative AI architecture deployable in a local or sovereign environment (open-source LLMs, pgvector vector indexing, no dependency on external APIs).
- Integration of reliability mechanisms: hallucination detection, verification of answer faithfulness to sources (LLM-judge), confidence thresholds and answer refusal.
- Security by design: secret encryption, access control (Row Level Security), multi-tenant isolation, rate limiting.
- Study of resilience and robustness: routing across multiple models, degraded mode on unavailability, comparison of cost / latency / quality trade-offs.
- Work on answer explainability: systematic source citation and decision justification for business users.

Automated decellularization system – Université de Sherbrooke

2025 : Canada : Univ. of Sherbrooke

- Development of an automated embedded system for the decellularization of biological tissue.
- Design of a Python software interface for bioreactor control.
- Integration and control of sensors, pumps and valves via microcontrollers.
- Embedded code optimization for real-time management of perfusion cycles.
- Experimental validation of the system on biological tissue.

Research on ROS2 architecture for chassis software prototyping

2024 – 2025 : France : Ampere Software Technology

- Study and development of an embedded ROS2 infrastructure for software prototyping on NUC platforms.
- Manual build and integration of ROS2 binaries in the embedded environment.
- Generation and deployment of ROS nodes from MATLAB models.
- Functional validation of several control modules in a vehicle environment.
- Contribution to improving the performance and reliability of the embedded ROS2 system.

EDUCATION

Master of Science in Artificial Intelligence
Data Science and Artificial Intelligence track

Epitech (France)
2026 - 2028

Advanced training in data science and artificial intelligence: generative AI, large language models (LLMs), RAG architectures and autonomous multi-agent systems, applied to concrete industrial use cases, alongside an apprenticeship at Yieloo.

Engineering Degree (Diplôme d'ingénieur)
Specialization in Embedded Software & Cybersecurity

ESEO - Grande École of engineering (France)
2023 - 2026

Advanced training in embedded software and cybersecurity, focused on real-time system design, communications security and the development of reliable solutions for industrial environments.

Research programme
Embedded systems & AI

Université de Sherbrooke (Canada)
2025

Participation in research programmes at the Université de Sherbrooke on the design of embedded systems dedicated to automated decellularization.

Training in management, marketing and entrepreneurial development

Stratégie Escape (France)
2024 - 2025

Skills in brand strategy, digital communication and commercial performance management, with a focus on innovation and business management.

Preparatory classes for engineering schools
Mathematics and Physics (CPGE)

Intensive two-year programme in mathematics and physics, focused on analysis, problem solving and preparation for engineering school entrance exams.

École Polytechnique
(Benin)
2021 - 2023

PERSONAL QUALITIES AND LEADERSHIP

Ambassador – International Relations & Integration

ESEO
2024 - 2025

- Selected by ESEO's International Relations Department to serve as a key point of contact for international students.
- Welcoming and integrating international students: supporting them with administrative steps and in finding their bearings at the school.
- Intercultural communication and promotion of exchanges within the student community, to foster inclusion.

President – Local Youth Foundation

ONG JPA ESEO
2020 - 2023

- Organizing community events and managing volunteer teams.
- Developing skills in communication, planning and decision-making.
- Strong ability to mobilize and unite people around collective projects.

Coach – U12 Football Club

FC - Espoir
2019 - 2021

- Supervising and training young players in a sporting and educational setting.
- Demonstrating creativity, adaptability and team spirit.
- Ability to work under pressure and motivate a group toward a common goal.
- Developing stress management and interpersonal communication skills.

CLUBS & ASSOCIATIONS

ESEO Angers
2023 - 2026

- Member of the ESEO Angers Robot club
- Member of the ESEO Angers automotive club (ESE'Auto)
- Member of the ESEO Angers football club

NOTES

Confidential projects are not presented, in order to respect confidentiality and non-disclosure commitments.

REFERENCES

Renaud DEBORNE – Ampere Software Technology (France)
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Kassoum SANOGO – tbmaestro (Canada)
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Sport **Travel** **Humanitarian**